A Crash Course in Forecast Reconciliation

杨杨卓然
Yangzhuoran Fin Yang

27 March 2026

About Me

Yangzhuoran Fin Yang

Assistant professor in Statistical Learning
Department of Data Analytics and Digitalisation
Maastricht University, the Netherlands

- Time series forecasting
- Outlier dectection
- R packages

yangzhuoranyang.com
yangzhuoran.yang@maastrichtuniversity.nl

Outline

1 Hierarchical and grouped time series

2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

5 Parallel and extensions

Outline

1 Hierarchical and grouped time series

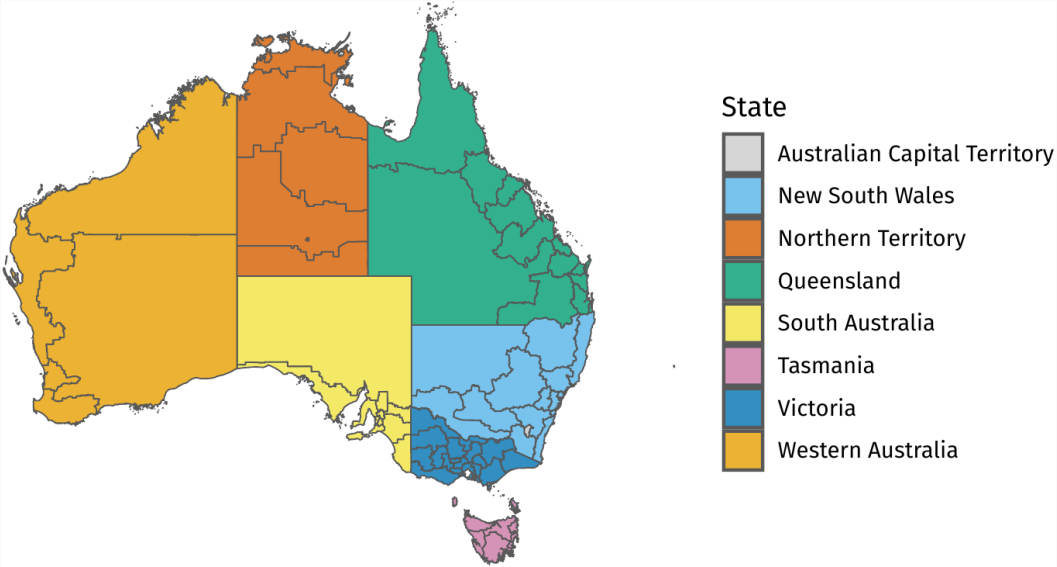
2 Traditional approaches

3 Forecast reconciliation

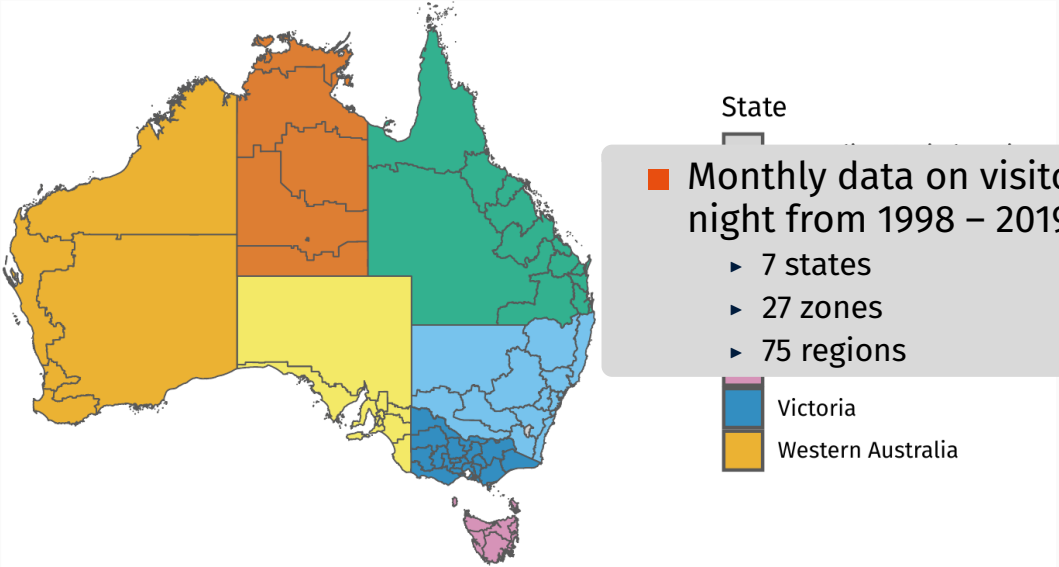
4 Projection and geometry

5 Parallel and extensions

Australian tourism regions

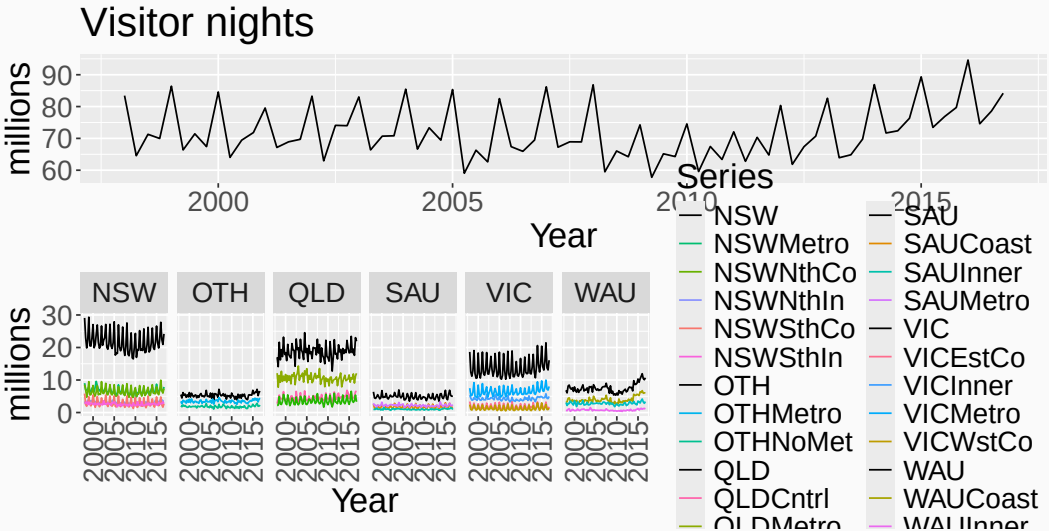


Australian tourism regions



- Monthly data on visitor night from 1998 – 2019
  - 7 states
  - 27 zones
  - 75 regions

Australian tourism data

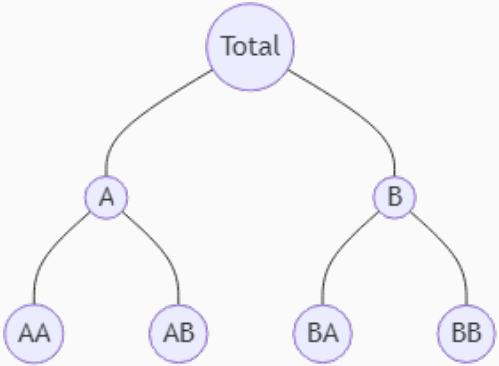


Australian tourism data

```
tsibble::tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:       Region, State, Purpose [304]
   Quarter Region   State           Purpose  Trips
     <qtr> <chr>    <chr>           <chr>    <dbl>
 1 1998 Q1 Adelaide South Australia Business  135.
 2 1998 Q2 Adelaide South Australia Business  110.
 3 1998 Q3 Adelaide South Australia Business  166.
 4 1998 Q4 Adelaide South Australia Business  127.
 5 1999 Q1 Adelaide South Australia Business  137.
 6 1999 Q2 Adelaide South Australia Business  200.
 7 1999 Q3 Adelaide South Australia Business  169.
 8 1999 Q4 Adelaide South Australia Business  134.
 9 2000 Q1 Adelaide South Australia Business  154.
10 2000 Q2 Adelaide South Australia Business  169.
# i 24,310 more rows
```

Hierarchical time series

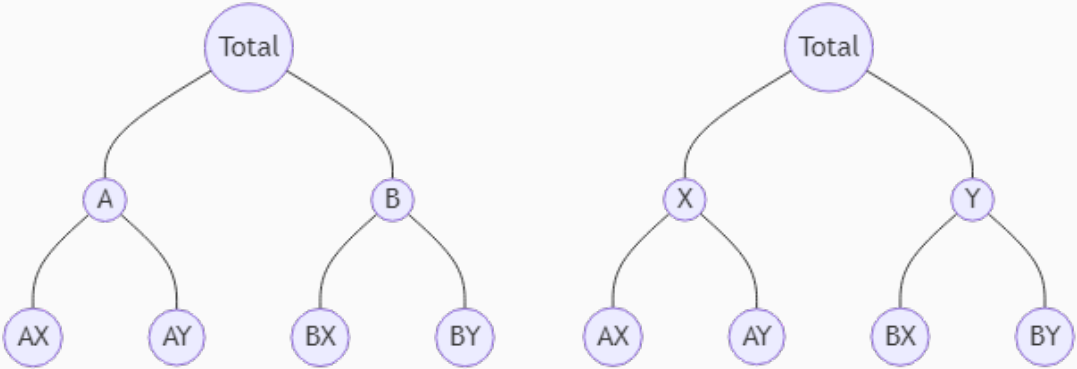


$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

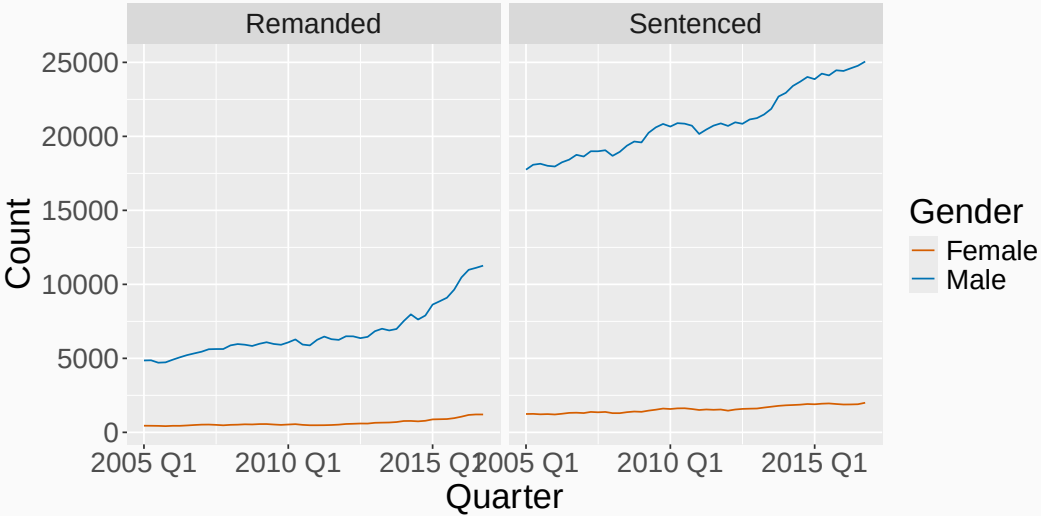
Grouped time series



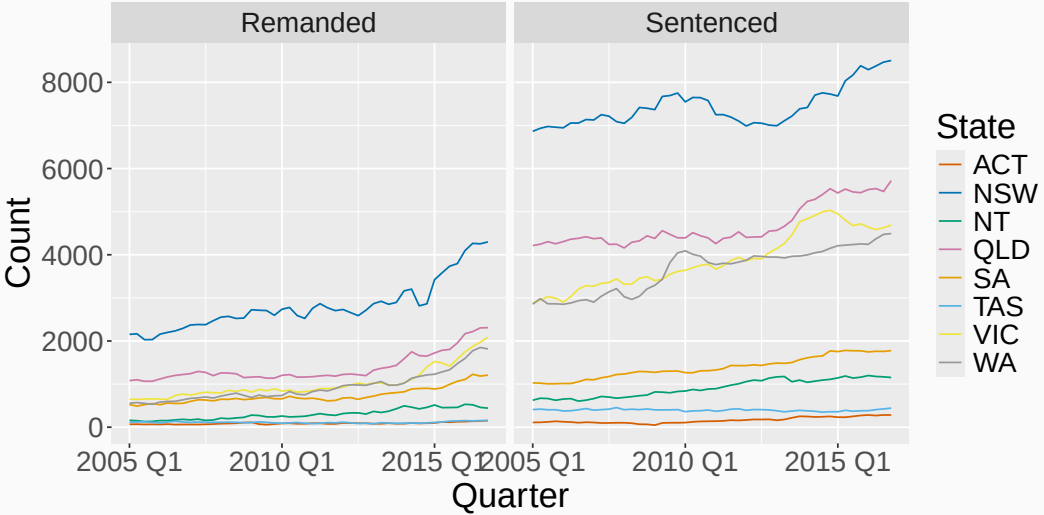
Australian prison population

```
# A tsibble: 3,072 x 6 [1Q]
# Key:       State, Gender, Legal, Indigenous [64]
   Quarter State Gender Legal    Indigenous Count
     <qtr> <chr> <chr>  <chr>    <chr>      <dbl>
 1 2005 Q1 ACT   Female Remanded ATSI           0
 2 2005 Q2 ACT   Female Remanded ATSI           1
 3 2005 Q3 ACT   Female Remanded ATSI           0
 4 2005 Q4 ACT   Female Remanded ATSI           0
 5 2006 Q1 ACT   Female Remanded ATSI           1
 6 2006 Q2 ACT   Female Remanded ATSI           1
 7 2006 Q3 ACT   Female Remanded ATSI           1
 8 2006 Q4 ACT   Female Remanded ATSI           0
 9 2007 Q1 ACT   Female Remanded ATSI           0
10 2007 Q2 ACT   Female Remanded ATSI           1
# i 3,062 more rows
```

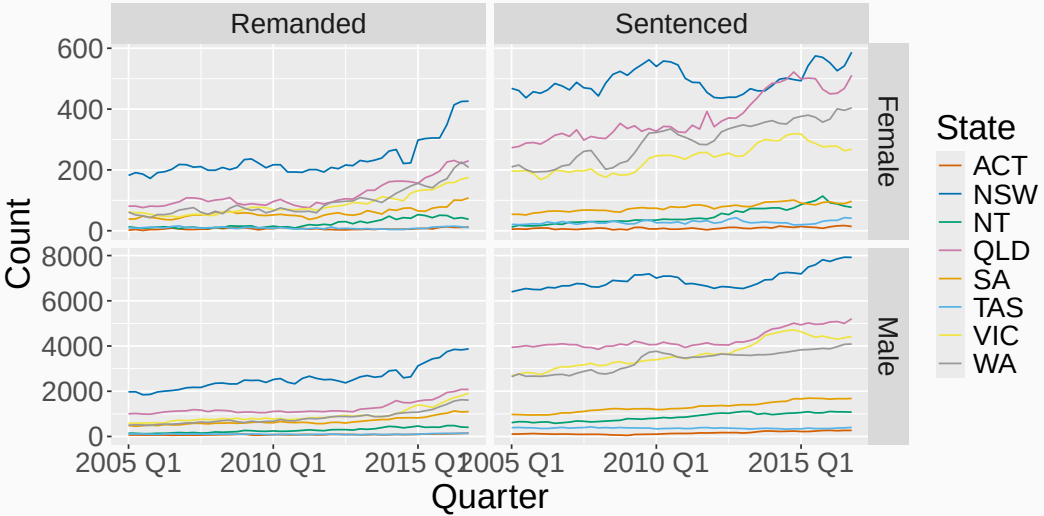
Australian prison population



Australian prison population



Australian prison population

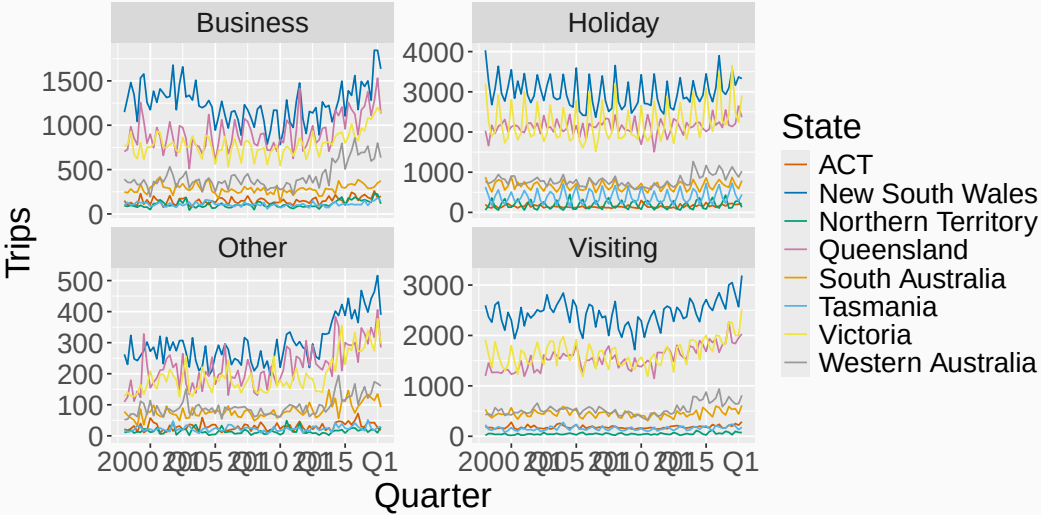


Mixed hierarchical and grouped structure

```
tsibble::tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:       Region, State, Purpose [304]
   Quarter Region   State           Purpose  Trips
     <qtr> <chr>    <chr>           <chr>    <dbl>
 1 1998 Q1 Adelaide South Australia Business  135.
 2 1998 Q2 Adelaide South Australia Business  110.
 3 1998 Q3 Adelaide South Australia Business  166.
 4 1998 Q4 Adelaide South Australia Business  127.
 5 1999 Q1 Adelaide South Australia Business  137.
 6 1999 Q2 Adelaide South Australia Business  200.
 7 1999 Q3 Adelaide South Australia Business  169.
 8 1999 Q4 Adelaide South Australia Business  134.
 9 2000 Q1 Adelaide South Australia Business  154.
10 2000 Q2 Adelaide South Australia Business  169.
# i 24,310 more rows
```

Travel purpose



Outline

1 Hierarchical and grouped time series

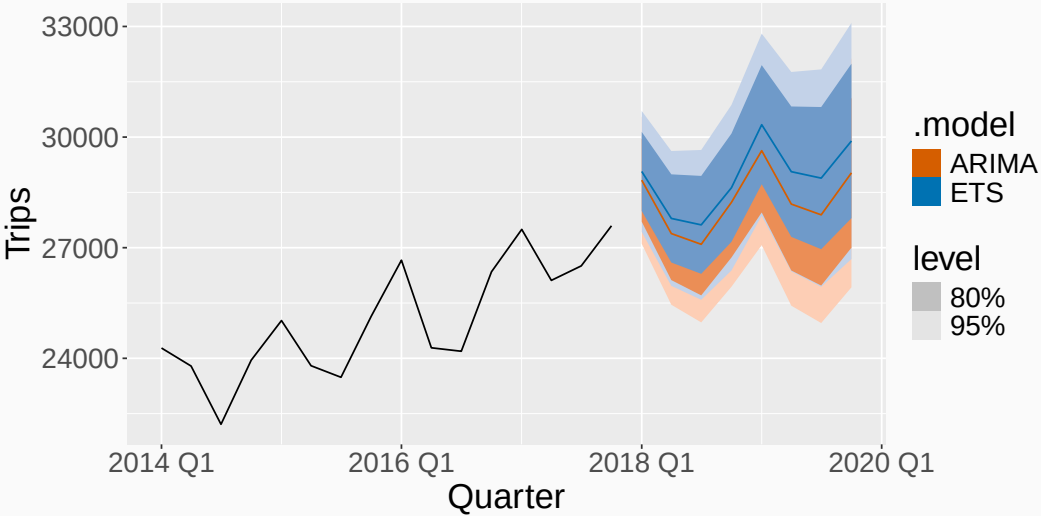
2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

5 Parallel and extensions

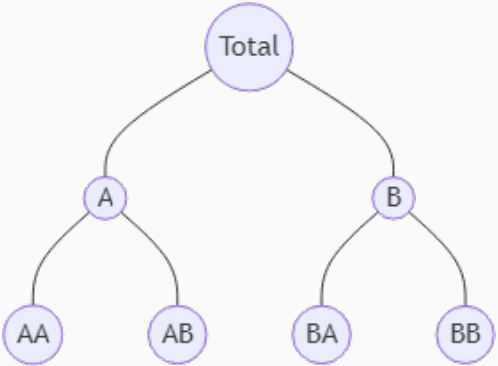
Forecasting method recap



Single level approaches

Goal

Obtain forecasts of every series at all levels, but they need to add up based on the hierarchical structure (**"coherent"**).

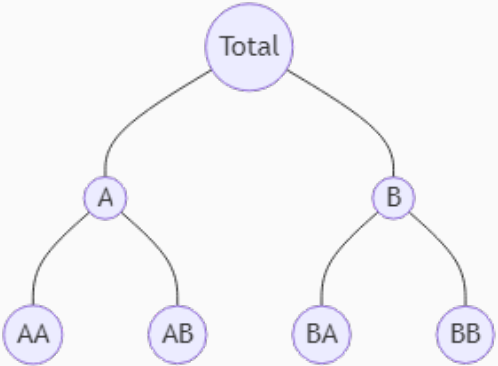


- Bottom-up

Single level approaches

Goal

Obtain forecasts of every series at all levels, but they need to add up based on the hierarchical structure (**"coherent"**).

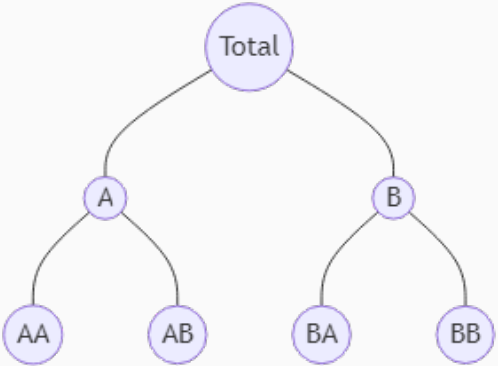


- Bottom-up
- Top-down

Single level approaches

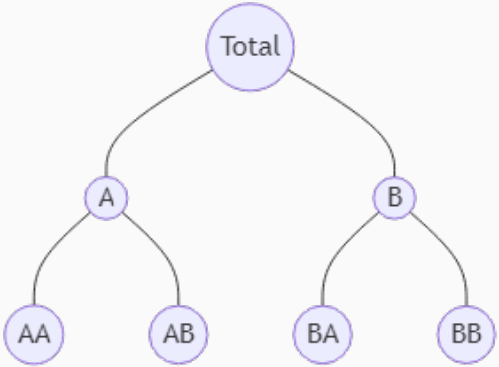
Goal

Obtain forecasts of every series at all levels, but they need to add up based on the hierarchical structure (**"coherent"**).



- Bottom-up
- Top-down
- Middle-out

Bottom-up

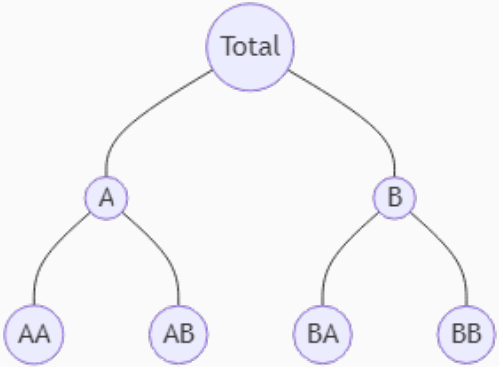


$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

Bottom-up



$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

$$\tilde{y}_t = \hat{y}_{AA,h} + \hat{y}_{AB,h} + \hat{y}_{BA,h} + \hat{y}_{BB,h}$$

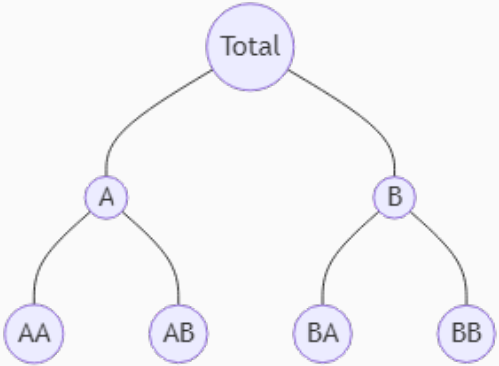
$$\tilde{y}_{A,h} = \hat{y}_{AA,h} + \hat{y}_{AB,h}$$

$$\tilde{y}_{B,h} = \hat{y}_{BA,h} + \hat{y}_{BB,h}$$

Terminology

- $y_{AA,t}$ and friends: the **bottom** level series.
- y_t the **top** level series.
- $y_{AA,t}$: The historical value of series AA at time t.
- $\hat{y}_{AA,h}$: The h-step-ahead forecast for series AA from a forecast model, potentially independently from other series. The **base** forecast.
- $\tilde{y}_{A,h}$: The **reconciled** h-step-ahead forecast for series A that is **coherent**.

Top-down

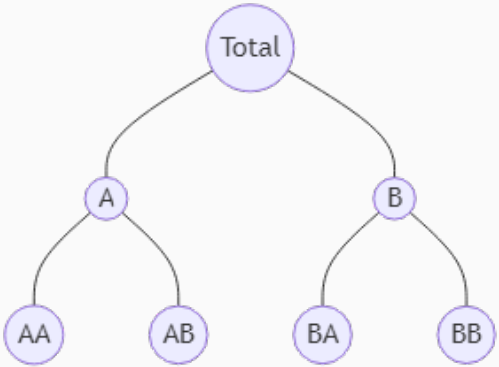


$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

Top-down



$$\tilde{y}_{AA,h} = p_{AA}\tilde{y}_t \quad \tilde{y}_{AB,h} = p_{AB}\tilde{y}_t$$

$$\tilde{y}_{BA,h} = p_{BA}\tilde{y}_t \quad \tilde{y}_{BB,h} = p_{BB}\tilde{y}_t$$

$$\tilde{y}_{A,h} = \tilde{y}_{AA,h} + \tilde{y}_{AB,h}$$

$$\tilde{y}_{B,h} = \tilde{y}_{BA,h} + \tilde{y}_{BB,h}$$

What are the proportions?

Average historical proportions

$$p_j = \frac{1}{T}\sum_{t=1}^{T}\frac{y_{j,t}}{y_t}$$

Proportions of the historical averages

$$p_j = \sum_{t=1}^{T}\frac{y_{j,t}}{T}\Big/\sum_{t=1}^{T}\frac{y_t}{T}$$

Forecast proportions

Use the proportion of the base forecasts at a lower level to the aggregate of the base forecasts at this level.

Middle-out

1. Choose a middle level to generate base forecasts
2. Bottom-up upwards
3. Top-down downwards

Middle-out

1. Choose a middle level to generate base forecasts
2. Bottom-up upwards
3. Top-down downwards

Problem

Seems like a waste - only one level is chosen (top, bottom or middle), and the information in other series are not used.

Outline

1 Hierarchical and grouped time series

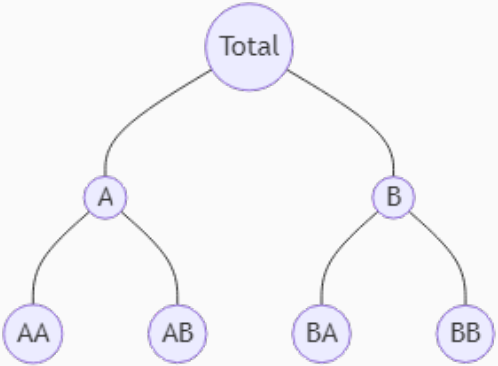
2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

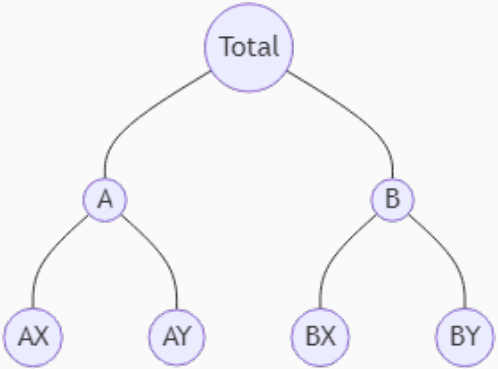
5 Parallel and extensions

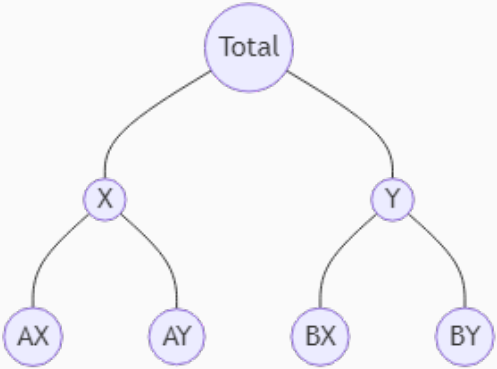
Matrix notation



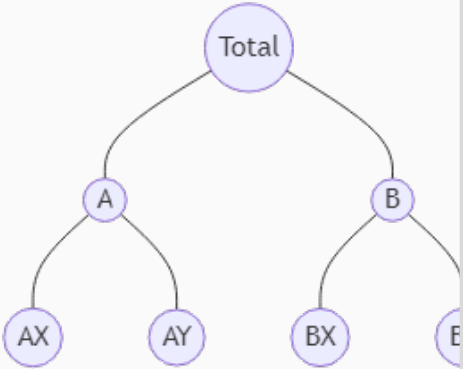
$$\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix}$$

Matrix notation



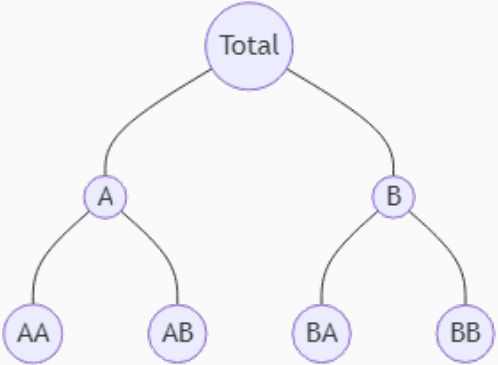


Matrix notation



$$
\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{X,t} \\ y_{Y,t} \\ y_{AX,t} \\ y_{AY,t} \\ y_{BX,t} \\ y_{BY,t} \end{bmatrix}
=
\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} y_{AX,t} \\ y_{AY,t} \\ y_{BX,t} \\ y_{BY,t} \end{bmatrix}
$$

Matrix notation



$$\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix}$$

$$\boldsymbol{y}_t = \boldsymbol{S}\boldsymbol{b}_t = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{b}_t$$

h-step-ahead forecast at time t

$$\boldsymbol{y}_t = \begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} \qquad \hat{\boldsymbol{y}}_{t+h} = \begin{bmatrix} \hat{y}_{t+h} \\ \hat{y}_{A,t+h} \\ \hat{y}_{B,t+h} \\ \hat{y}_{AA,t+h} \\ \hat{y}_{AB,t+h} \\ \hat{y}_{BA,t+h} \\ \hat{y}_{BB,t+h} \end{bmatrix} \qquad \hat{\boldsymbol{b}}_{t+h} = \begin{bmatrix} \hat{y}_{AA,t+h} \\ \hat{y}_{AB,t+h} \\ \hat{y}_{BA,t+h} \\ \hat{y}_{BB,t+h} \end{bmatrix} \qquad \tilde{\boldsymbol{y}}_{t+h} = \begin{bmatrix} \tilde{y}_{t+h} \\ \tilde{y}_{A,t+h} \\ \tilde{y}_{B,t+h} \\ \tilde{y}_{AA,t+h} \\ \tilde{y}_{AB,t+h} \\ \tilde{y}_{BA,t+h} \\ \tilde{y}_{BB,t+h} \end{bmatrix}$$

Forecast reconciliation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{M}\hat{\boldsymbol{y}}_{t+h}$$

Bottom-up

$$\boldsymbol{G} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Top-down

$$\boldsymbol{G} = \begin{bmatrix} p_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_4 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Forecast reconciliation

Limitation

The traditional methods uses only a single level with limited information.

Forecast reconciliation

Limitation

The traditional methods uses only a single level with limited information.

Goal

Finding the *"optimal"* $\boldsymbol{G}$ matrix to give the most accurate coherent forecasts in.

Forecast reconciliation

Limitation

The traditional methods uses only a single level with limited information.

Goal

Finding the *"optimal"* $\boldsymbol{G}$ matrix to give the most accurate coherent forecasts in.

Optimality

Optimal in the sense of smallest **forecast error variance** while maintaining **unbiasedness**.

Unbiasedness

If the base forecasts are **unbiased** in the sense that

$$\mathrm{E}(\hat{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t),$$

where $\mathcal{I}_t = \boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_t$ denote data observed up to time t,

we want the reconciled forecast to maintain this property such that

$$\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t).$$

Unbiasedness

Goal

$$\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t).$$

$$\begin{aligned}\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) &= \mathrm{E}(\boldsymbol{SG}\hat{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \boldsymbol{SG}\,\mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t)\\ &= \boldsymbol{SG}\,\mathrm{E}(\boldsymbol{Sb}_{t+h}|\mathcal{I}_t) = \boldsymbol{SGS}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t)\end{aligned}$$

Unbiasedness

Goal

$$\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t).$$

$$\begin{aligned}\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) &= \mathrm{E}(\boldsymbol{SG}\hat{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \boldsymbol{SG}\,\mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t)\\ &= \boldsymbol{SG}\,\mathrm{E}(\boldsymbol{Sb}_{t+h}|\mathcal{I}_t) = \boldsymbol{SGS}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t)\end{aligned}$$

Conclusion

When the base forecasts are unbiased, the reconciled forecasts will be unbiased if and only if $\boldsymbol{SGS} = \boldsymbol{S}$, or equivalently, $\boldsymbol{GS} = \boldsymbol{I}$.

Forecast error variance

The base forecast error is $\hat{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h}$ and the base forecast error variance is $\text{Var}(\hat{\varepsilon}_{t+h}|\mathcal{I}_t) = \boldsymbol{W}_h$:

$$\boldsymbol{W}_h = \begin{pmatrix} \text{Var}(\hat{\varepsilon}_{t+h}) & \text{Cov}(\hat{\varepsilon}_{t+h}, \hat{\varepsilon}_{A,t+h}) & \cdots & \text{Cov}(\hat{\varepsilon}_{t+h}, \hat{\varepsilon}_{BB,t+h}) \\ \text{Cov}(\hat{\varepsilon}_{A,t+h}, \hat{\varepsilon}_{t+h}) & \text{Var}(\hat{\varepsilon}_{A,t+h}) & \cdots & \text{Cov}(\hat{\varepsilon}_{A,t+h}, \hat{\varepsilon}_{BB,t+h}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\varepsilon}_{BB,t+h}, \hat{\varepsilon}_{t+h}) & \text{Cov}(\hat{\varepsilon}_{BB,t+h}, \hat{\varepsilon}_{A,t+h}) & \cdots & \text{Var}(\hat{\varepsilon}_{BB,t+h}) \end{pmatrix}$$

Forecast error variance

Define

$$\hat{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h} \text{ and } \tilde{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h}$$

Goal

We want to find the forecast error variance $\mathrm{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t)$.

Forecast error variance

Define

$$\hat{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h} \text{ and } \tilde{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h}$$

Goal

We want to find the forecast error variance $\text{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t)$.

$$\begin{aligned}
\tilde{\varepsilon}_{t+h} - \hat{\varepsilon}_{t+h} &= \boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h} - \boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h} \\
\tilde{\varepsilon}_{t+h} &= \hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\hat{\boldsymbol{y}}_{t+h} \\
\tilde{\varepsilon}_{t+h} &= \hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})(\boldsymbol{y}_{t+h} - \hat{\varepsilon}_{t+h}) \\
\tilde{\varepsilon}_{t+h} &= \boldsymbol{SG}\hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\boldsymbol{y}_{t+h}
\end{aligned}$$

$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\boldsymbol{S}\boldsymbol{b}_{t+h}$$

$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h},$$

because $\boldsymbol{SGS} = \boldsymbol{S}$ to maintain unbiasedness.

$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\boldsymbol{S}\boldsymbol{b}_{t+h}$$
$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h},$$

because $\boldsymbol{SGS} = \boldsymbol{S}$ to maintain unbiasedness.

Forecast error variance

The reconciled forecast error variance is

$$\text{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t) = \text{Var}(\boldsymbol{SG}\hat{\varepsilon}_{t+h}|\mathcal{I}_t) = \boldsymbol{SG}\,\text{Var}(\hat{\varepsilon}_{t+h}|\mathcal{I}_t)\boldsymbol{G}'\boldsymbol{S}' = \boldsymbol{SGW}_h\boldsymbol{G}'\boldsymbol{S}'.$$

Minimising sum of forecast error variances

Trace of a matrix

The trace of a square matrix is the sum of the diagnal elements of that matrix.

Minimising sum of forecast error variances

Trace of a matrix

The trace of a square matrix is the sum of the diagnal elements of that matrix.

The MinT optimal reconciliation approach

Wickramasuriya et al. (2019) propose the **MinT** method to minimise the trace of the forecast error varianve $\mathrm{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t)$:

$$\begin{aligned}\boldsymbol{G} &= \arg\min \mathrm{tr}(\boldsymbol{S}\boldsymbol{G}\boldsymbol{W}_h\boldsymbol{G}'\boldsymbol{S}') \text{ s.t. } \boldsymbol{S}\boldsymbol{G}\boldsymbol{S} = \boldsymbol{S} \\ &= (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}\end{aligned}$$

A proof can be found with Ando and Narita (2024).

Minimising ~~sum of~~ forecast error variances

In fact, the MinT solution not only reaches the minimum of the *trace* - the *sum* of the forecast error variances, but also reaches the minimum forecast error varianve individually for each series (Yang et al in a secret working paper).

Alternative representation

Structural representation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{M}\hat{\boldsymbol{y}}_{t+h}$$

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

Mapping matrix:

$$\boldsymbol{M} = \boldsymbol{S}(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

Alternative representation

Structural representation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{M}\hat{\boldsymbol{y}}_{t+h}$$

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

Mapping matrix:

$$\boldsymbol{M} = \boldsymbol{S}(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

Zero constrained representation

$$\boldsymbol{y}_t = \boldsymbol{S}\boldsymbol{b}_t = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{b}_t$$

Constraint matrix:

$$\boldsymbol{C} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \end{bmatrix}$$

$$\boldsymbol{C}\boldsymbol{y}_t = \boldsymbol{0}$$

$$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{W}_h\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{W}\boldsymbol{C}')^{-1}\boldsymbol{C}.$$

Zero constrained representation

THe structural representation

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

requires solving inverse of $\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S}$ with the dimension being **the number of series at the bottom level**.

Zero constrained representation

THe structural representation

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

requires solving inverse of $\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S}$ with the dimension being **the number of series at the bottom level**. The zero constrained representation (see Di Fonzo and Girolimetto 2023) is **mostly more efficient**.

$$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{W}_h\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{W}\boldsymbol{C}')^{-1}\boldsymbol{C}$$

requires solving inverse of $\boldsymbol{C}\boldsymbol{W}\boldsymbol{C}'$ with the dimension being **the number of series of the upper levels**.

Forecast error variance covariance estimation

To perform forecast reconciliation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h},$$

we need

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1},$$

which in turn requires $\boldsymbol{W}_h$.

Problem

How do we get the base forecast error variance $\boldsymbol{W}_h$?

1. OLS

Set $\boldsymbol{W}_h = k_h \boldsymbol{I}$ for all h, where $k_h > 0$. This gives the OLS estimator.

The value of k_h does not matter because it cancells out in $(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$.

1. OLS

Set $\boldsymbol{W}_h = k_h \boldsymbol{I}$ for all h, where $k_h > 0$. This gives the OLS estimator.

The value of k_h does not matter because it cancells out in $(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$.

- Fast, independent from the data
- Distance reducing
- Good empirical performance
- Effectively says the base forecast error variances are all k_hs and covariances are all 0s.

2. WLS (variance scaling)

Set $\boldsymbol{W}_h = k_h \mathrm{diag}(\hat{\boldsymbol{W}}_1)$ for all h, where $k_h > 0$,

$$\hat{\boldsymbol{W}}_1 = \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_t'.$$

2. WLS (variance scaling)

Set $\boldsymbol{W}_h = k_h \mathrm{diag}(\hat{\boldsymbol{W}}_1)$ for all h, where $k_h > 0$,

$$\hat{\boldsymbol{W}}_1 = \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_t'.$$

- Scales the base forecasts using the variance of the residuals
- The WLS estimator using *variance scaling*.
- Assumes base forecast errors are uncorrelated between each other.

3. Structural scaling

Set $\boldsymbol{W}_h = k_h \Lambda$ for all h, where $k_h > 0$, $\Lambda = \mathrm{diag}(\boldsymbol{S}\mathbf{1})$, and $\mathbf{1}$ is a unit vector.

3. Structural scaling

Set $\boldsymbol{W}_h = k_h \Lambda$ for all h, where $k_h > 0$, $\Lambda = \mathrm{diag}(\boldsymbol{S}\mathbf{1})$, and $\mathbf{1}$ is a unit vector.

- Only depends on the hierarchical structure.
- Useful when residuals are not available.

3. Structural scaling

Set $\boldsymbol{W}_h = k_h \Lambda$ for all h, where $k_h > 0$, $\Lambda = \text{diag}(\boldsymbol{S}\mathbf{1})$, and $\mathbf{1}$ is a unit vector.

- Only depends on the hierarchical structure.
- Useful when residuals are not available.
- Assumes bottom-level base forecast errors each have variance k_h and are uncorrelated between each other.
- Each element of Λ contains the number of forecast error variances contributing to each node.

4a. Sample covariance

Set $\boldsymbol{W}_h = k_h \hat{\boldsymbol{W}}_1$ for all h, where $k_h > 0$.

- Direct estimation.
- Assumes only that the error covariance are proportional to each other.

4a. Sample covariance

Set $\boldsymbol{W}_h = k_h \hat{\boldsymbol{W}}_1$ for all h, where $k_h > 0$.

- Direct estimation.
- Assumes only that the error covariance are proportional to each other.
- Difficult when the structure is large but series are short.

4a. Sample covariance

Set $\boldsymbol{W}_h = k_h \hat{\boldsymbol{W}}_1$ for all h, where $k_h > 0$.

- Direct estimation.
- Assumes only that the error covariance are proportional to each other.
- Difficult when the structure is large but series are short.

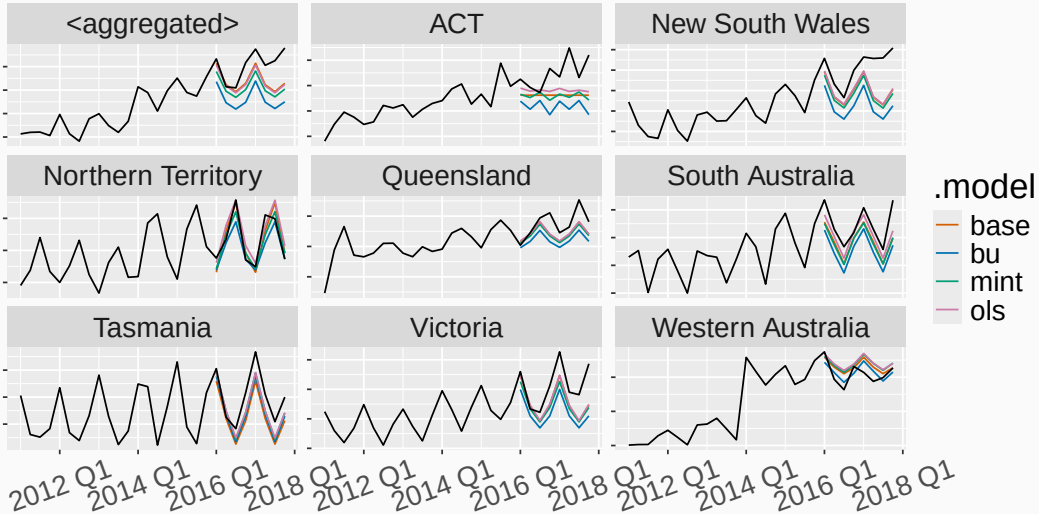
4b. Shrinkage estimator

Shrinking covariance towards zero (Schäfer and Strimmer 2005) and/or **shrinking variance** towards their median (Opgen-Rhein and Strimmer 2007).

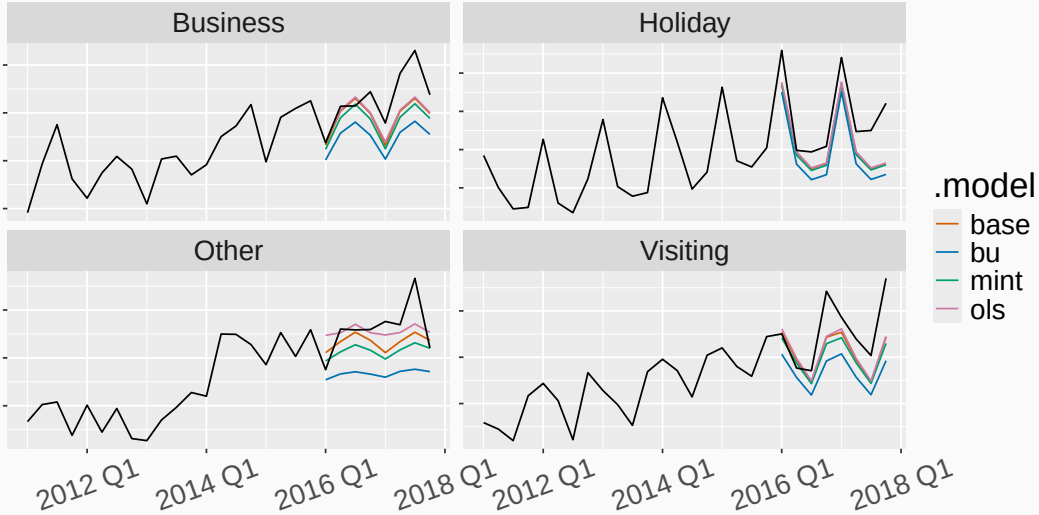
- Positive definite, and
- Numerically stable.

`fable` in R supports shrinking covariance.

Forecast Australian domestic tourism



Forecast Australian domestic tourism



Outline

1 Hierarchical and grouped time series

2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

5 Parallel and extensions

Reference

The main reference of this section is Panagiotelis et al. (2021).

OLS recap

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{u}_i$$

$$\boldsymbol{y} = \hat{\beta}_0 + \hat{\beta}_1 \boldsymbol{x} + \hat{\boldsymbol{u}}$$

$$\boldsymbol{y} = \boldsymbol{X}\hat{\beta} + \hat{\boldsymbol{u}}$$

$$\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

OLS recap

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{u}_i$$

$$\boldsymbol{y} = \hat{\beta}_0 + \hat{\beta}_1 \boldsymbol{x} + \hat{\boldsymbol{u}}$$

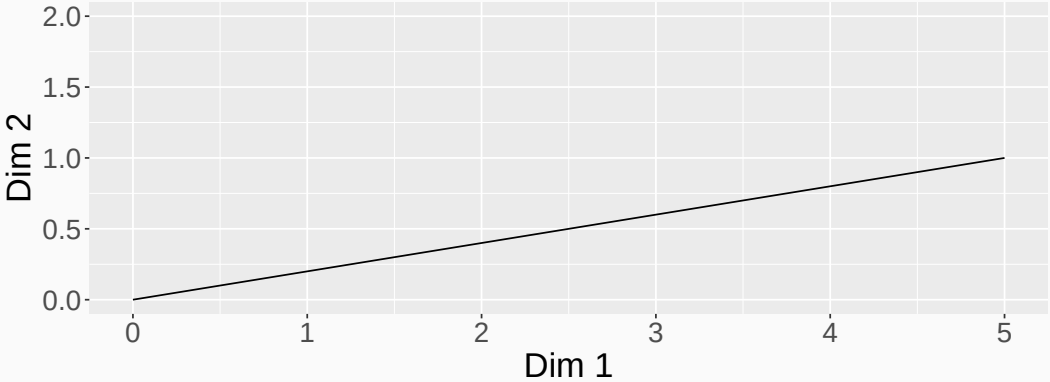
$$\boldsymbol{y} = \boldsymbol{X}\hat{\beta} + \hat{\boldsymbol{u}}$$

$$\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

$$\begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \hat{\beta}_0 + \begin{bmatrix} 5 \\ 1 \end{bmatrix} \hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix} 1 & 5 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} + \hat{\boldsymbol{u}}$$

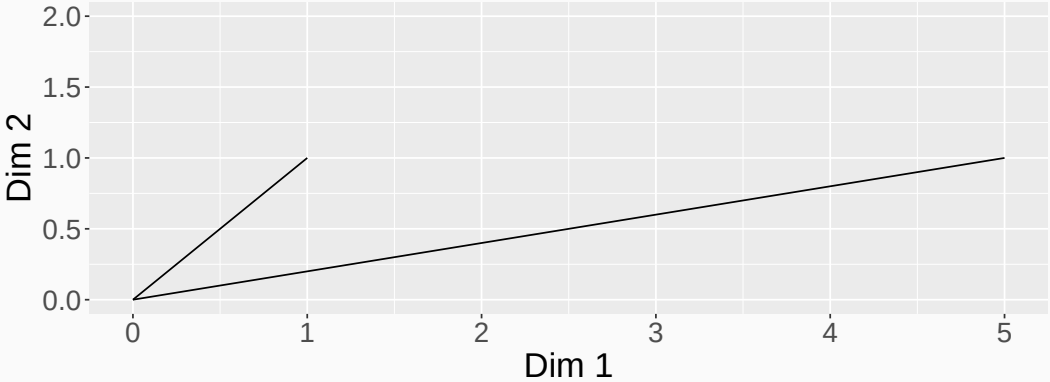
OLS geometry - 2 observations

$$\begin{bmatrix}5\\2\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



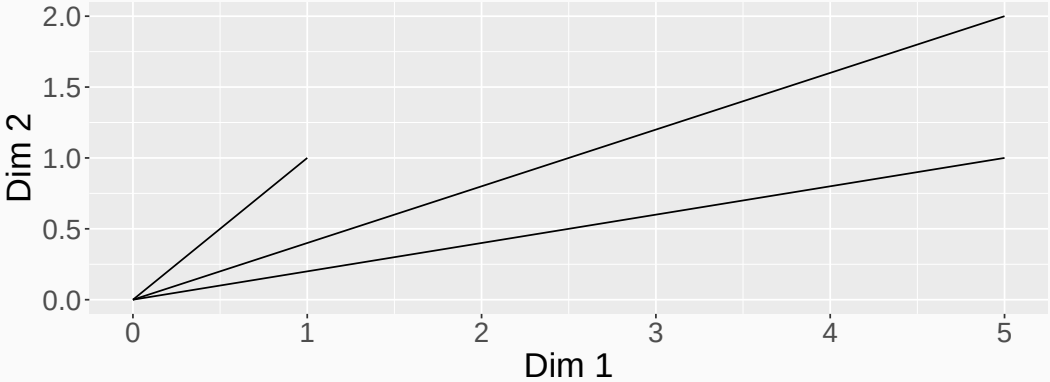
OLS geometry - 2 observations

$$\begin{bmatrix}5\\2\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



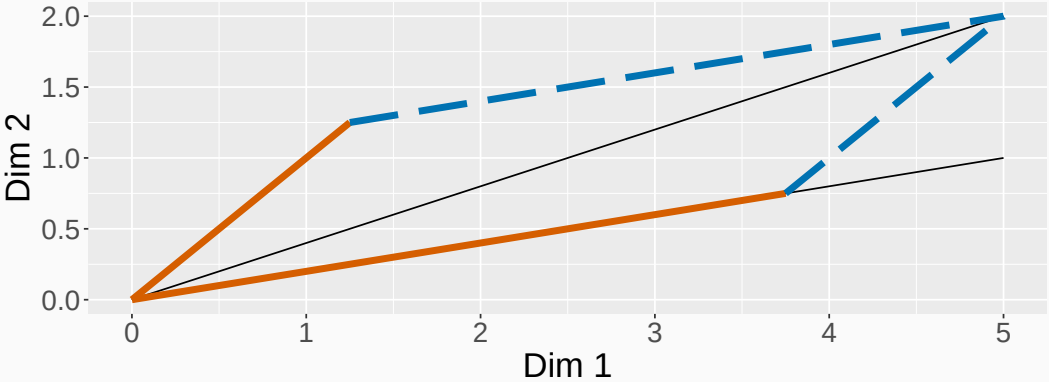
OLS geometry - 2 observations

$$\begin{bmatrix}5\\2\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



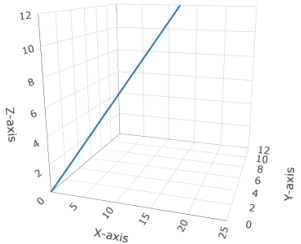
OLS geometry - 2 observations

$$\begin{bmatrix}5\\2\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



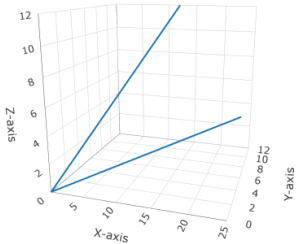
OLS geometry - 3 observations

$$\begin{bmatrix}5\\2\\3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



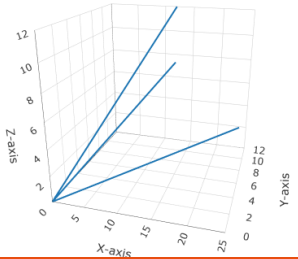
OLS geometry - 3 observations

$$\begin{bmatrix}5\\2\\3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



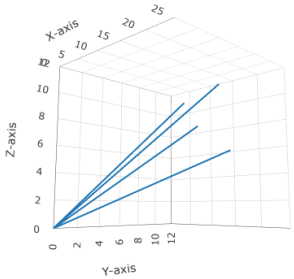
OLS geometry - 3 observations

$$\begin{bmatrix}5\\2\\3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



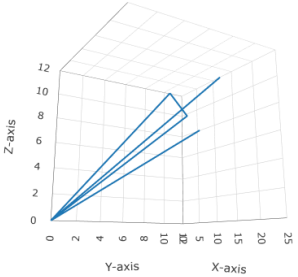
OLS geometry - 3 observations

$$\begin{bmatrix}5\\2\\3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



OLS geometry - 3 observations

$$\begin{bmatrix}5\\2\\3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



Forecast reconciliation as regression

We can write forecast reconciliation as

$$\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\beta_{t+h} + \boldsymbol{u}_{t+h}$$

where $\beta_{t+h} = \mathrm{E}(\boldsymbol{b}_{T+h}|\mathcal{I}_t)$.

Forecast reconciliation as regression

We can write forecast reconciliation as

$$\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\beta_{t+h} + \boldsymbol{u}_{t+h}$$

where $\beta_{t+h} = \mathrm{E}(\boldsymbol{b}_{T+h}|\mathcal{I}_t)$. The OLS solution of this regression is

$$\hat{\beta}_{t+h} = (\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\hat{\boldsymbol{y}}_{t+h},$$

which gives

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'$$

corresponding to the OLS estimator for the forecast error variance from before.

Forecast reconciliation

Coherent subspace

THe linear subspace $\mathfrak{s}$ for which the hierarchical structure hold for all $\boldsymbol{y} \in \mathfrak{s}$.

Forecast reconciliation

Coherent subspace

THe linear subspace $\mathfrak{s}$ for which the hierarchical structure hold for all $\boldsymbol{y} \in \mathfrak{s}$.

Coherent point forecasts

A vector of point forecast $\check{\boldsymbol{y}}_{t+h}$ of all series in the hierarchy is *coherent* if $\check{\boldsymbol{y}}_{t+h} \in \mathfrak{s}$.

Forecast reconciliation

Coherent subspace

THe linear subspace $\mathfrak{s}$ for which the hierarchical structure hold for all $\boldsymbol{y} \in \mathfrak{s}$.

Coherent point forecasts

A vector of point forecast $\check{\boldsymbol{y}}_{t+h}$ of all series in the hierarchy is *coherent* if $\check{\boldsymbol{y}}_{t+h} \in \mathfrak{s}$.

OLS forecast reconciliation

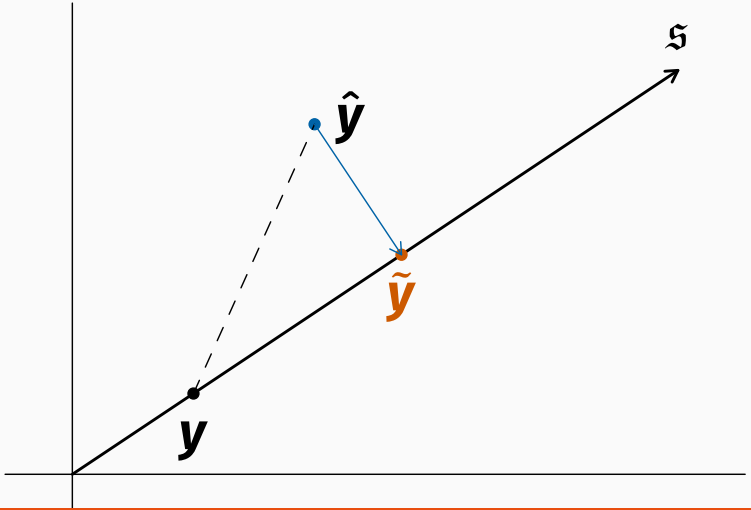
The OLS solution is an orthogonal projection of the base forecasts onto the coherent subspace $\mathfrak{s}$.

Distance-Reducing Property

If $\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{SG}\hat{\boldsymbol{y}}_{t+h}$, where $\boldsymbol{G}$ is such that $\boldsymbol{SG}$ is an orthogonal projection onto $\mathfrak{s}$, then

$$||\boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h}|| \leq ||\boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h}||.$$

Orthogonal projection



Oblique projection - MinT

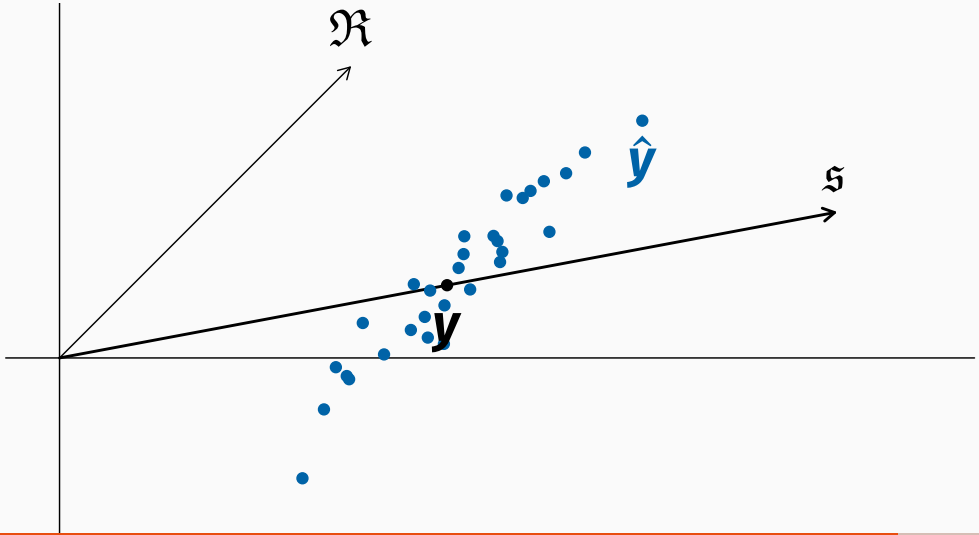
The MinT method is not an orthogonal projection. It is an oblique one.

MinT

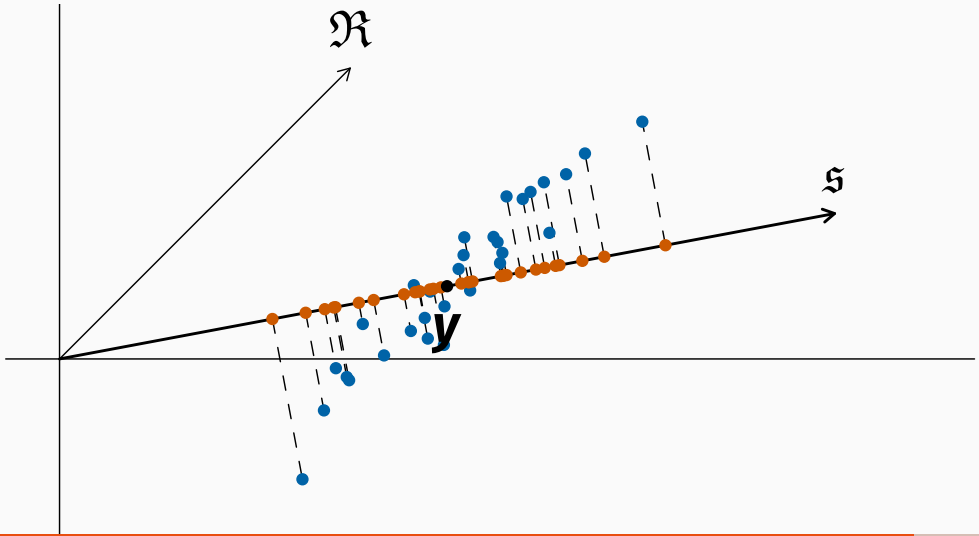
$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

While orthogonal projection always improve on base forecasts, MinT minimises the loss function in **expectation** - minimum forecast error variances.

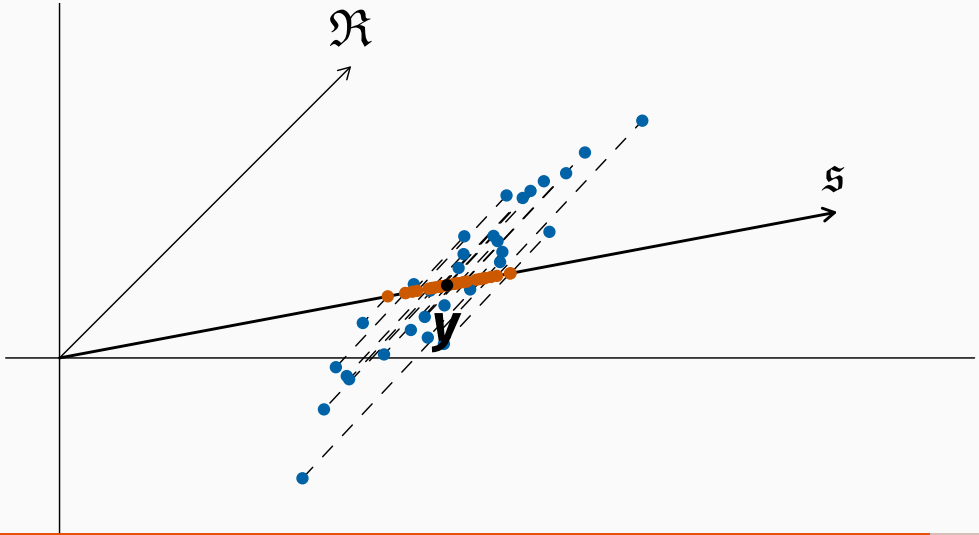
Base forecasts



Orthogonal projection



Oblique projection - MinT



Outline

1 Hierarchical and grouped time series

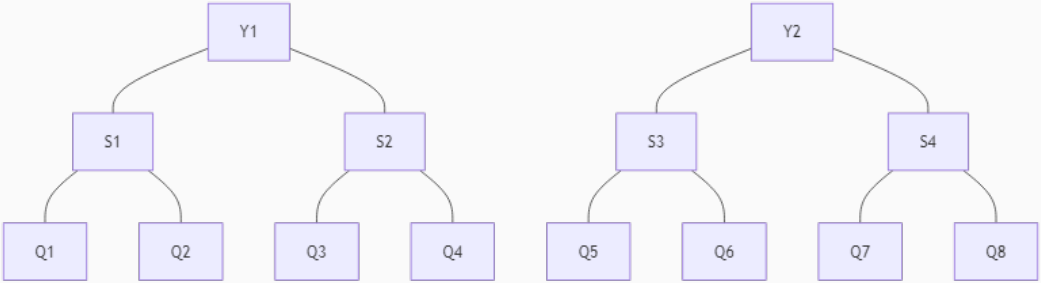
2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

5 Parallel and extensions

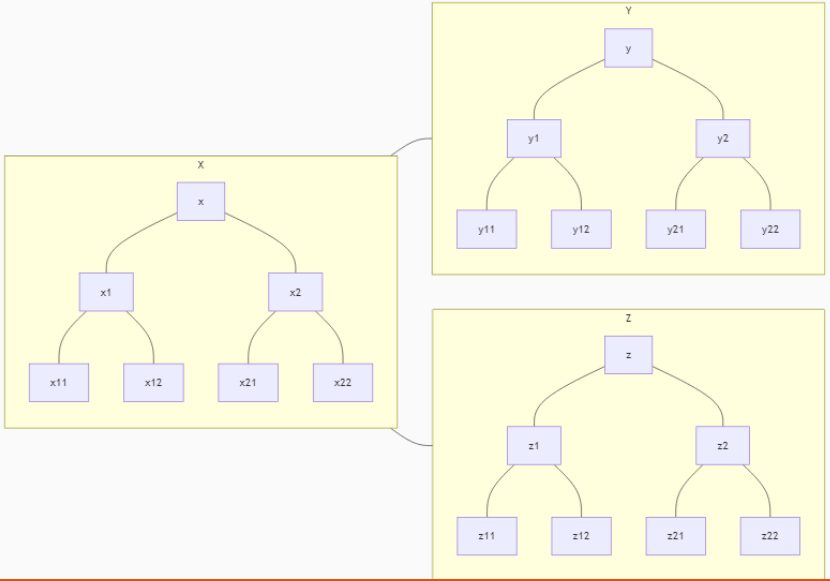
Temporal forecast reconciliation



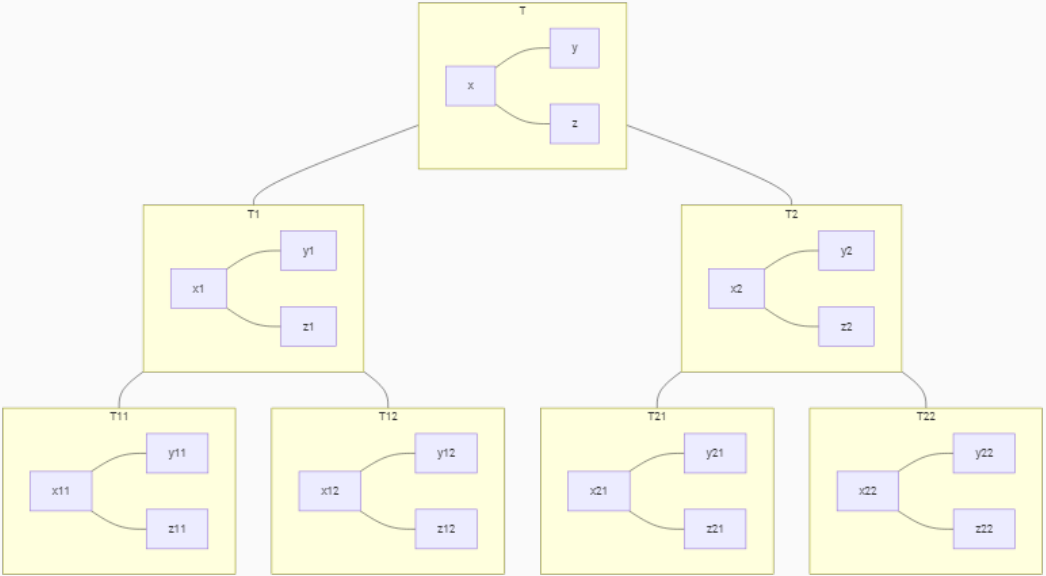
Temporal forecast reconciliation

- Single series observed at different frequencies
- Structural representation and zero constrained representation both still hold
- Very underused
  - Can be applied in most of the case as an extra step to improve accuracy
- See Athanasopoulos et al. (2017).

Cross-temporal forecast reconciliation



Cross-temporal forecast reconciliation



Cross-temporal forecast reconciliation

- Combining the two different hierarchies.
- More difficult to solve.
- Different algorithms and heuristics.
- See Di Fonzo and Girolimetto (2023)

Probabilistic forecast reconciliation

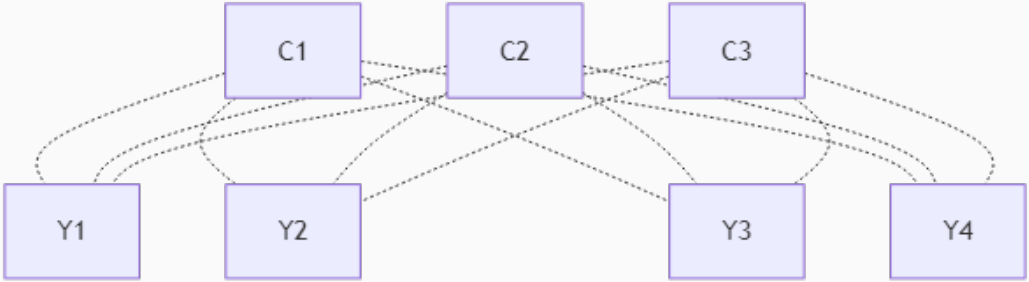
Forecast distribution

Apart from forecasting a single point, we can have a full forecast distribution.

Probabilistic forecast reconciliation

- Any possible forecasts that do not correctly add up according to the hierarchical structure are assigned zero probability.
- See Panagiotelis et al. (2023).

Forecast Linear Augmented Projection (FLAP)



FLAP

- Artificial upper level series
- Suitable for series even without a natural hierarchy
- Theoretical guarantee to reduce forecast error variance
  - The more artificial components the better
  - Optimal solution given a set of components
- Good empirical performance
- See Yang et al. (2024)

References i

Ando, Sakai, and Futoshi Narita. 2024. "An Alternative Proof of Minimum Trace Reconciliation." *Forecasting* 6 (2): 456–61. https://doi.org/10.3390/forecast6020025.

Athanasopoulos, George, Rob J Hyndman, Nikolaos Kourentzes, and Fotios Petropoulos. 2017. "Forecasting with temporal hierarchies." *Eur. J. Oper. Res.* 262 (1): 60–74. https://doi.org/10.1016/j.ejor.2017.02.046.

Di Fonzo, Tommaso, and Daniele Girolimetto. 2023. "Cross-temporal forecast reconciliation: Optimal combination method and heuristic alternatives." *Int. J. Forecast.* 39 (1): 39–57. https://doi.org/10.1016/j.ijforecast.2021.08.004.

Opgen-Rhein, Rainer, and Korbinian Strimmer. 2007. "Accurate ranking of differentially expressed genes by a distribution-free shrinkage approach." *Stat. Appl. Genet. Mol. Biol.* 6 (1). https://doi.org/10.2202/1544-6115.1252.

Panagiotelis, Anastasios, George Athanasopoulos, Puwasala Gamakumara, and Rob J Hyndman. 2021. "Forecast reconciliation: A geometric view with new insights on bias correction." *Int. J. Forecast.* 37 (1): 343–59. https://doi.org/10.1016/j.ijforecast.2020.06.004.

References ii

Panagiotelis, Anastasios, Puwasala Gamakumara, George Athanasopoulos, and Rob J Hyndman. 2023. "Probabilistic forecast reconciliation: Properties, evaluation and score optimisation." *Eur. J. Oper. Res.* 306 (2): 693–706. https://doi.org/10.1016/j.ejor.2022.07.040.

Schäfer, Juliane, and Korbinian Strimmer. 2005. "A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics." *Stat. Appl. Genet. Mol. Biol.* 4 (1). https://doi.org/10.2202/1544-6115.1175.

Wickramasuriya, Shanika L, George Athanasopoulos, and Rob J Hyndman. 2019. "Optimal forecast reconciliation for hierarchical and grouped time series through trace minimization." *J. Am. Stat. Assoc.* 114 (526): 804–19. https://doi.org/10.1080/01621459.2018.1448825.

Yang, Yangzhuoran Fin, George Athanasopoulos, Rob J Hyndman, and Anastasios Panagiotelis. 2024. "Forecast Linear Augmented Projection (FLAP): A free lunch to reduce forecast error variance." *arXiv [Stat.ME].* http://arxiv.org/abs/2407.01868.